

MEADOW LAKE, SK, Canada

WMO: 719700

Lat: 54.1286N

Lon: 108.5110W

Elev: 481

StdP: 95.68

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.3	-32.9	-40.2	0.1	-36.1	-36.8	0.1	-32.8	11.7	-4.5	10.1	-4.9	1.8	250	0.703

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	28.3	18.3	26.5	17.6	24.8	16.8	20.0	26.1	18.9	24.6	17.8	23.1	4.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.0	13.7	22.7	16.7	12.6	21.4	15.7	11.8	20.3	59.5	26.0	55.7	24.6	52.3	23.1	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.9	8.7	7.8	-40.6	31.9	3.7	2.0	-43.3	33.3	-45.4	34.5	-47.5	35.6	-50.2	37.1
			-40.7	22.5	3.7	1.5	-43.3	23.6	-45.5	24.5	-47.5	25.3	-50.2	26.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	1.2	-16.4	-14.4	-7.2	2.7	9.8	14.6	16.9	15.5	10.5	3.0	-6.7	-14.3	(6)
(7)		DBStd	13.48	9.46	8.06	8.18	5.71	4.69	3.36	2.81	3.30	4.32	4.86	6.77	8.32	(7)
(8)		HDD10.0	3849	817	682	534	226	63	4	0	2	45	221	501	753	(8)
(9)		HDD18.3	6272	1076	916	793	470	267	119	61	97	237	474	751	1012	(9)
(10)		CDD10.0	655	0	0	0	6	56	142	215	173	59	4	0	0	(10)
(11)		CDD18.3	36	0	0	0	0	1	6	17	10	1	0	0	0	(11)
(12)		CDH23.3	718	0	0	0	3	59	123	266	208	56	2	0	0	(12)
(13)		CDH26.7	147	0	0	0	1	12	24	56	46	10	0	0	0	(13)
(14)	Wind	WSAvg	3.5	3.4	3.3	3.6	3.9	4.0	3.7	3.3	3.2	3.6	3.7	3.5	3.3	(14)
(15)	Precipitation	PrecAvg	467	21	14	17	27	36	88	84	68	43	30	20	19	(15)
(16)		PrecMax	664	40	46	40	102	95	175	178	176	107	100	53	51	(16)
(17)		PrecMin	327	6	2	2	0	2	37	26	18	4	6	0	5	(17)
(18)		PrecStd	92	10	9	9	21	25	38	34	35	27	19	13	9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.1	4.6	11.1	22.7	28.2	29.3	30.8	30.7	28.1	21.1	9.6	5.1	(19)
MCWB			1.7	1.9	5.6	11.4	14.9	17.7	20.2	19.2	16.9	12.9	5.5	2.3	(20)	
2%		DB	2.3	1.8	6.9	18.6	24.8	26.5	28.1	27.6	24.8	16.4	6.3	2.3	(21)	
		MCWB	0.7	-0.2	3.1	9.4	13.5	16.6	19.3	18.5	15.6	9.7	3.5	0.4	(22)	
5%		DB	0.1	-0.5	4.7	15.4	22.3	24.4	26.2	25.6	21.8	13.5	4.0	0.0	(23)	
		MCWB	-1.2	-1.7	1.9	7.8	12.1	15.4	18.7	17.8	14.1	8.1	1.8	-1.3	(24)	
10%		DB	-2.8	-2.5	3.0	12.4	19.9	22.5	24.5	23.7	19.1	10.8	1.9	-2.4	(25)	
		MCWB	-3.7	-3.5	0.8	6.4	11.0	14.8	17.7	16.9	12.7	6.7	0.3	-3.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.1	2.0	6.2	12.1	16.1	19.7	22.2	21.1	18.2	13.2	6.0	2.4	(27)
MCDWB			3.7	4.2	10.8	21.1	24.5	26.1	27.6	27.4	25.6	20.6	9.0	4.4	(28)	
2%		WB	0.8	0.1	3.4	9.9	14.5	18.1	20.6	19.5	16.2	10.5	3.9	0.6	(29)	
		MCDWB	2.3	1.6	6.4	17.5	22.3	23.4	26.0	25.7	23.1	14.9	6.0	2.2	(30)	
5%		WB	-1.2	-1.7	2.1	8.2	13.1	16.9	19.4	18.4	14.7	8.6	1.9	-1.3	(31)	
		MCDWB	0.2	-0.6	4.4	14.6	20.5	22.4	25.0	24.3	20.8	12.8	3.7	0.0	(32)	
10%		WB	-3.7	-3.6	1.0	6.7	11.8	15.7	18.3	17.3	13.3	7.1	0.4	-3.4	(33)	
		MCDWB	-2.8	-2.5	2.8	11.9	18.6	20.7	23.4	22.6	18.2	10.4	1.8	-2.4	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.6	11.9	11.7	11.9	14.2	12.4	12.7	13.2	13.0	10.7	9.0	10.1	(35)
MCDBR			11.5	12.3	12.0	17.2	18.5	16.4	16.1	17.3	18.5	15.9	11.4	11.7	(36)	
MCWBR			10.0	10.7	8.9	9.8	9.2	8.1	8.5	9.2	10.2	9.8	8.6	10.3	(37)	
5% WB		MCDBR	11.4	11.8	11.3	16.3	16.1	13.5	14.6	15.2	17.2	14.9	10.7	11.5	(38)	
		MCWBR	10.0	10.4	8.6	9.7	8.5	7.5	8.3	8.7	9.9	9.5	8.3	10.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub		0.242	0.244	0.292	0.320	0.345	0.365	0.369	0.361	0.311	0.286	0.250	0.217	(40)
(41)		taud		2.328	2.371	2.353	2.431	2.403	2.367	2.377	2.383	2.525	2.560	2.469	2.365	(41)
(42)		Ebn at Noon		748	880	895	913	901	883	871	857	870	822	746	723	(42)
(43)		Edh at Noon		53	72	93	100	109	114	111	104	79	61	47	41	(43)
(44)	All-Sky Solar Radiation	RadAvg		0.88	1.93	3.46	4.67	5.69	5.72	5.87	4.79	3.35	1.91	0.86	0.62	(44)
(45)		RadStd		0.09	0.13	0.24	0.38	0.50	0.42	0.38	0.38	0.38	0.32	0.12	0.06	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page